RG 2

The interaction term: try to integrate $\varphi_>$ perturbatively

$$\hat{u} = \frac{u}{4}\int dx\, \varphi^4(x) = \frac{u}{4}\int \frac{d\vec{q}_1 d\vec{q}_2 d\vec{q}_3 d\vec{q}_4}{(2\pi)^{4d}} (2\pi)^d \delta(\vec{q}_1+\vec{q}_2+\vec{q}_3+\vec{q}_4)$$

$$(\varphi_<(q_1)+\varphi_>(q_2))(\varphi_<(q_2)+\varphi_>(q_2))(\varphi_<(q_3)+\varphi_>(q_3))(\varphi_<(q_4)+\varphi_>(q_4))$$

$$Z = \int D\varphi_< \exp\left\{-\int_0^{\Lambda/b} \frac{d\vec{q}}{(2\pi)^d} \frac{t+\rho q^2}{2} |\varphi_<(\vec{q})|^2\right\} \cdot \left\langle e^{-\hat{u}(\varphi_<,\varphi_>)}\right\rangle_{\varphi_>}$$

where $\langle \hat{O}\rangle_{\varphi_>} = \frac{1}{Z_>}\int D\varphi_> \hat{O}\, e^{-\int_{\Lambda/b}^{\Lambda} \frac{d\vec{q}}{(2\pi)^d} \frac{t+\rho q^2}{2} |\varphi_>(q)|^2}$

So that we obtain $Z = \int D\varphi_< e^{-S_<}$

where $S_< = \int \frac{d\vec{q}}{(2\pi)^d} \frac{t+\rho q^2}{2} |\varphi_<|^2 - \ln\left\langle e^{-\hat{u}(\varphi_<,\varphi_>)}\right\rangle_{\varphi_>}$

The last term generates <u>cumulant</u> expansion

$$\ln\langle e^{-\hat{u}}\rangle_{\varphi_>} = \ln\left\langle 1 - \hat{u} + \frac{1}{2}\hat{u}^2 + \dots\right\rangle_{\varphi_>} =$$

$$= \ln\left[1 - \langle\hat{u}\rangle_{\varphi_>} + \frac{1}{2}\langle\hat{u}^2\rangle_{\varphi_>} + \dots\right] = \left(\ln(1-x) = -x - \frac{1}{2}x^2 + \dots\right)$$

$$\cong -\langle\hat{u}\rangle + \frac{1}{2}\langle\hat{u}^2\rangle - \frac{1}{2}\left(\langle\hat{u}\rangle + \frac{1}{2}\langle\hat{u}^2\rangle\right)^2 =$$

$$= \underbrace{-\langle\hat{u}\rangle}_{\substack{1^{st}\text{ order}\\ \text{in } u.}} + \frac{1}{2}\underbrace{\left(\langle\hat{u}^2\rangle - \langle\hat{u}\rangle^2\right)}_{2^{nd}\text{ order in } u.} + \dots$$

Since action $S_>$ of the fast modes $\varphi_>$ is gaussian, the average in $\langle U \rangle_{\varphi_>}$ can involve only 0, 2, or 4 fields $\varphi_>$.

* Term with "0" $\varphi_>$: $\frac{u}{4}\int_0^{\Lambda/b} \frac{d\vec{q}_1 \dots d\vec{q}_4}{(2\pi)^{4d}} (2\pi)^d \delta(q_1+q_2+q_3+q_4) \cdot$

$\cdot\ \varphi_<(q_1)\, \varphi_<(q_2)\, \varphi_<(q_3)\, \varphi_<(q_4)$ ← this is renormalized u

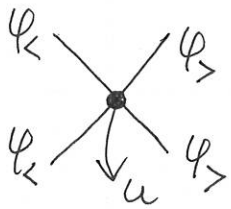
* Term with "4" $\varphi_>$: $\frac{u}{4}\int \frac{d\bar{q}_1 \dots d\bar{q}_4}{(2\pi)^{4d}} (2\pi)^d \delta(q_1+q_2+q_3+q_4)$

$\cdot\ \varphi_>(q_1)\, \varphi_>(q_2)\, \varphi_>(q_3)\, \varphi_>(q_4)$

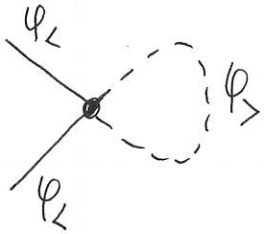
this term has no dependence on $\varphi_<$ at all!

* Terms with "2" powers of $\varphi_>$:

Graphic representation



Coefficient 4×3



$$3u \int \frac{d\bar{q}_1\, d\bar{q}_2\, d\bar{q}_3\, d\bar{q}_4}{(2\pi)^{3d}} \delta(q_1+q_2+q_3+q_4) \cdot$$

$$\cdot\ \varphi_<(q_1)\, \varphi_<(q_2) \underbrace{\langle \varphi_>(q_3)\, \varphi_>(q_4) \rangle}_{\frac{(2\pi)^d \delta(q_3+q_4)}{t+p q_3^2}} =$$

$$= 3u \int \frac{d\bar{q}_1\, d\bar{q}_2}{(2\pi)^d} \varphi_<(q_1)\, \varphi_<(q_2)\, \delta(q_1+q_2) \cdot \underbrace{\int_{\frac{\Lambda}{b}}^{\Lambda} \frac{d\bar{q}_3}{(2\pi)^d} \frac{1}{t+p q_3^2}}_{\text{contraction of fast modes}}$$

$\left[0 < q_{1,2} < \frac{\Lambda}{b}\right]$

Thus, to first order in $\underline{\underline{u}}$

$$S_< = \int \frac{d\vec{q}}{(2\pi)^d} \frac{1}{2}\Big(t + \rho q^2 + \underbrace{6u \int \frac{d\vec{q}_3}{(2\pi)^d} \frac{1}{t+\rho q_3^2}}_{\text{correction to } t}\Big) |\varphi_<(q)|^2$$

$$+ \frac{1}{4} u \int \frac{dq_1 \dots dq_3}{(2\pi)^{3d}} \varphi_<(q_1)\, \varphi_<(q_2)\, \varphi_<(q_3)\, \varphi_<(-q_1-q_2-q_3)$$

$$\left[0 < q_{1,2,3} < \frac{\Lambda}{b}\right]$$

Now we need to <u>rescale</u> momenta $\vec{q} = b^{-1}\vec{q}\,'$

and <u>renormalize</u> fields $\varphi_<\left(\frac{q'}{b}\right) = z\,\varphi'(q')$

$$S_< = \int_0^\Lambda \frac{d\vec{q}\,'}{(2\pi)^d} b^{-d} z^2 \frac{\tilde{t} + \rho q'^2 b^{-2}}{z} |\varphi'(q')|^2$$

$$+ \frac{1}{4} u\, b^{-3d} z^4 \int_0^\Lambda \frac{d\vec{q}_1'\, d\vec{q}_2'\, d\vec{q}_3'}{(2\pi)^{3d}} \varphi'(q_1')\, \varphi'(q_2')\, \varphi'(q_3')\, \varphi'(-q_1'-q_2'-q_3')$$

Thus new parameters are

$$\begin{cases} t' = b^{-d} z^2 \tilde{t} & \to b^2 \tilde{t} \\ \rho' = b^{-d-2} z^2 \rho & \to \text{choose } z = b^{1+\frac{d}{2}} \text{ so that } \rho' = \rho \\ u' = b^{-3d} z^4 u & \to b^{-3d+4(\frac{d}{2}+1)} u = b^{4-d} u \end{cases}$$

In the vicinity of $t=0$, $u=0$:

$$\begin{cases} t_b' = b^2\left(t + 6u \int_{N_b}^{\Lambda} \frac{dk}{(2\pi)^d} \frac{1}{\rho k^2 + t}\right) \\ u_b' = b^{4-d} u \end{cases}$$

With $b = 1+\delta l$, the range of k-integration is very narrow

<u>In $d=3$</u>

$$\int_{\frac{\Lambda}{b}}^{\Lambda} \frac{dk\, k^2 \cdot 4\pi}{(2\pi)^3} \frac{1}{t+pk^2} = \frac{1}{2\pi^2} \frac{\Lambda^2}{t+p\Lambda^2} \underbrace{\left(\Lambda - \frac{\Lambda}{b}\right)}_{\int_{\Lambda/b}^{\Lambda} dk} = \frac{\Lambda^3 \delta l}{2\pi^2(t+p\Lambda^2)}$$

$$\Lambda - \frac{\Lambda}{1+\delta l} = \Lambda - \Lambda(1-\delta l) = \Lambda \delta l$$

$$\Rightarrow \quad t'_b = t(1+\delta l) = t + \delta l \frac{dt}{dl} = (1+2\delta l)\left[t + 6u \frac{\Lambda^3 \delta l}{2\pi^2(t+p\Lambda^2)}\right] =$$

$$= t + 2t\,\delta l + \frac{6u\Lambda^3 \delta l}{2\pi^2(t+p\Lambda^2)} + O(\delta l^2)$$

$$\Rightarrow \quad \frac{dt}{dl} = 2t + \frac{3u\Lambda^2}{\pi^2(t+p\Lambda^2)}$$

And $\dfrac{du}{dl} = (4-d)\,u$

the fixed point is still at $t^* = 0$, $u^* = 0$

In <u>d-dimensions</u>

$$\int_{\frac{\Lambda}{b}}^{\Lambda} \frac{dk\, k^{d-1} S_d}{(2\pi)^d} \frac{1}{t+pk^2} = \frac{\Lambda^{d-1}}{t+p\Lambda^2} \frac{S_d}{(2\pi)^d} \Lambda \delta l =$$

spherical angle in d-dim.

$$= \frac{K_d \Lambda^d}{t+p\Lambda^2} \delta l$$

$$\Rightarrow \quad \frac{dt}{dl} = 2t + \frac{6u K_d \Lambda^d}{t+p\Lambda^2}$$

$$\frac{du}{dl} = (4-d)\,u$$

To second order in u (Ch. 5.6 of Kardar)
one generates $-Au^2$ in equation for $\frac{dt}{dl}$

and $-Bu^2$ in equation for $\frac{du}{dl}$ $\quad (A, B > 0)$

$$\Rightarrow \begin{cases} \frac{dt}{dl} = 2t + \frac{6u K_d \Lambda^d}{t + p\Lambda^2} - Au^2 \\ \frac{du}{dl} = (4-d)u - Bu^2 \end{cases}$$

$\Rightarrow$ now the fixed point $\frac{du^*}{dl} = 0$ is $(4-d)u^* = B(u^*)^2$

$$u^* = \frac{4-d}{B}$$

In order to ensure that u^* is small
(we obtain all series by expanding in $\underline{u}$!)
we need to require $\frac{4-d}{B} \ll 1$

$\Rightarrow$ consider dimension of space d as variable
and let $d = 4 - \varepsilon \Rightarrow 4 - d = \varepsilon \ll 1$

<u>The ε - expansion!</u>